
Causally-Faithful Radiology Report Generation via Dual-Adversarial Counterfactual Training

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Abstract

Vision-language models that draft radiology reports often produce text that does not actually reflect the underlying chest X-ray. Diagnostic studies have documented this evidence-blindness, and existing fixes wrap the trained model in inference-time filters or conformal selection without changing what the model has learned. We instead intervene at training time. We propose dual-adversarial causal-faithfulness training, a LoRA fine-tune of MedGemma-4B-IT with two ingredients: a symmetric adversarial filter that downweights training examples that a text-only or an image-only predictor can already solve, and a causal-faithfulness loss that penalises the model whenever its probability of the gold token fails to drop under image masking. We pair the recipe with IMG_correct, a refinement of the standard image-masking-gap diagnostic that excludes a known failure mode in which a consistency-loss-trained model merely flips its top prediction under masking without losing support for the right answer. On the public-data OpenI chest-X-ray collection, our method drives the image-masking gap from near zero to roughly 75 pp at matched token-level accuracy across three seeds. A complementary cross-counterfactual evaluation, which we run as an honesty check, finds the gain is largely specific to the counterfactual the loss trained on: image cropping and centre occlusion collapse the gap, so the model has tightly fit its training-time counterfactual rather than the image as a whole. Applying IMG_correct retroactively to a set of 18

previously-reported verifier configurations reveals that the methodological refinement verifies prior numbers rather than overturning them. We release training data, code, and weights.

1. Introduction

A growing body of work documents that multimodal medical AI systems — vision- language models for radiology, ophthalmology, and pathology — make their predictions in ways that disregard the visual input. Restrepo et al. (2025) introduce Selective Modality Shifting (SMS) on MIMIC-CXR and FairVLMed and show that six open-source VLMs, including those fine-tuned for medical use, exhibit a marked dependency on text input that persists despite the presence of complementary visual information. This phenomenon is the medical-imaging analogue of the partial-input shortcuts documented for natural language inference (Gururangan et al., 2018; Poliak et al., 2018), and the multimodal-reward-model shortcuts documented by Li et al. (2025a).

Two facts about the existing literature motivate the present work. First, evidence-blindness diagnostics in medical imaging have been measurement-only: Restrepo et al. (2025) explicitly leave training-time mitigation to future work, and the conformal-selection literature (Gui et al., 2024; Li et al., 2025b; Mohri & Hashimoto, 2024) similarly wraps an existing predictor without modifying the underlying model. Second, the only training-time mitigation applied at scale to multimodal models — the Shortcut-Failure Coefficient (SFC) reweighting of Li et al. (2025a) — operates only on the text-only side, despite the fact that an image-only shortcut is just as conceptually possible and, on naturalistic medical data, just as likely. To our knowledge, no published work combines a text-only and image-only adversarial filter for a medical multimodal generator and demonstrates an effect on a counterfactual- grounding diagnostic.

Our two primary contributions are as follows. First,

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we develop and release a fine-tuning recipe — dual-adversarial causal- faithfulness training — that addresses both training-distribution shortcuts symmetrically and adds an online causal-faithfulness loss with a token-wise correctness-gain margin. The full method is a LoRA fine-tune of MedGemma-4B-IT (Google HAI-DEF, 2025) trained on OpenI (Demner-Fushman et al., 2016) using only public-data sources. Second, we contribute a methodological patch to the standard image-masking gap (IMG): IMG_correct, which counts only those examples where the model’s correct-class probability dropped by a margin τ under image masking, ruling out the induced inversion failure mode in which a consistency-loss-trained model is wrong on the image-zeroed input by flipping to a different-but-still-wrong answer rather than dropping its support for the gold class. We apply IMG_correct retroactively to our prior verifier work and find that, contra the pre-registered concern, the induced-inversion contribution is uniformly small. As a third instrumentation release — not a validated contribution — we ship a training-time gradient-norm monitor; we explicitly do not claim a deployable monitor in the present paper because the held-out mech-interp sweep needed to validate its predictive AUROC was beyond our compute envelope.

A cross-counterfactual eval (§5.2) honestly reports that the headline +72.58 pp IMG_correct improvement does not generalise cleanly to localised counterfactuals (image-cropping, centre-occlusion); the model has tightly fit the trained per-channel-mean and noise distributions but does not transfer to counterfactuals it has not seen. We frame the contribution accordingly: this paper demonstrates that training-time counterfactual interventions can be tightly-fit on the in-distribution counterfactual, not that the resulting model is image-grounded in a deeper sense. A more diverse training-time counterfactual mix is the natural follow-up.

2. Related Work

Evidence-blindness diagnostics. Restrepo et al. (2025) introduce the Selective Modality Shifting (SMS) family of perturbation- based diagnostics that swap images or text between samples with opposing labels. Our IMG_correct is conceptually a refinement of the image-perturbation-only branch of SMS, with an additional gold-class- probability-drop margin that excludes induced inversion. Earlier diagnostic work for natural-language entailment partial-input baselines (Gururangan et al., 2018; Poliak et al., 2018) provides the conceptual precedent for treating shortcut-solvability as a property of the dataset rather

than the model.

Adversarial filtering for shortcuts. Li et al. (2025a) introduce the Shortcut-Failure Coefficient, a continuous training-sample reweighting scheme for multimodal reward models that down-weights samples solvable by a text-only model. The SFC is operated only along the text axis; the obvious complementary mechanism — an image-only shortcut detector that mirrors the text-only SFC — is not implemented in Li et al. (2025a) or, to our knowledge, any subsequent work for the medical-imaging setting. We extend the family with a symmetric dual filter aggregated as $\min(w_{\text{text}}, w_{\text{image}})$ to downweight rows that are unimodally solvable along either axis.

Conformal selection on top of frozen multimodal models. Gui et al. (2024) train an alignment predictor on a reference dataset of human-labelled alignment scores and apply a data-dependent threshold at deployment. Li et al. (2025b) treat each generated text claim as an individual hypothesis and demonstrate, on LLaVa-1.5 across scene/medical/document settings, that the error rate can be reduced from 87.8% to 10.0% at a 95.3% true-positive rate. Both methods operate on a frozen underlying model. Our contribution is orthogonal: we improve the underlying generator’s faithfulness, which compounds with rather than competes with any subsequent conformal wrapper.

Audit-and-repair frameworks for radiology generators. RadFlag (Zhang et al., 2025) samples reports at varying temperatures and uses self-agreement to flag hallucinated claims; ReXTrust (Hardy et al., 2025) trains a self-attention head over LVLm hidden states (AUROC 0.875); FactCheXcker (Heiman et al., 2025) de-hallucinates measurement statements via a code-generation pipeline. All three are post-hoc, inference-time hallucination filters operating on a frozen generator; none implements a training-time intervention against evidence-blindness. The training-time, model-modifying contribution proposed here stacks with each (a more faithful underlying generator delivers fewer hallucinations to filter regardless of the filtering recipe).

Methodological-audit precedent. Recht et al. (2019) re-audit ImageNet-trained classifiers under a held-out-from-the-original- distribution test set and find that aggregate accuracy claims do not transfer. Our IMG_correct audit of prior IMG numbers is the medical- imaging analogue: we accept that prior IMG numbers are correct under the IMG definition but ask how much of the reported gap survives under a more

demanding definition that excludes induced inversion.

3. Methods

3.1. Backbone and parameter-efficient fine-tuning

We fine-tune MedGemma-4B-IT (Google HAI-DEF, 2025), a 4.31 B-parameter image-text instruction-tuned variant of Gemma-3 trained on a mixture of medical images including chest radiographs, pathology slides, dermatology images, and fundus photography. The model exposes a standard AutoModelForImageTextToText interface with a `pixel_values` keyword argument that accepts a $(B, 3, H, W)$ tensor of normalised image patches. We wrap the base model with a LoRA adapter (Hu et al., 2022) of rank 16 and alpha 32, applied to the `q_proj/k_proj/v_proj/o_proj` modules of every transformer block. Trainable parameter count: 11.9 M (0.28% of base); the rest is frozen. CheXagent-2 (Chen et al., 2024) was the original target backbone but its tokenizer-embedded image paradigm does not accept a `pixel_values` keyword, blocking counterfactual training without substantial reverse-engineering effort. MAIRA-2 (Bannur et al., 2024) requires similar backbone-specific work that we did not undertake. Our backbone choice was driven by this engineering constraint.

3.2. Composite training loss

The training objective combines three components,

$$\mathcal{L}_{\text{total}}(x) = w(x) \cdot \mathcal{L}_{\text{SFT}}(x) + \lambda_{\text{faith}} \cdot \mathcal{L}_{\text{faith}}(x), \quad (1)$$

where $w(x) \in [w_{\min}, 1]$ is the per-row weight produced by the dual adversarial filter (§3.3), $\mathcal{L}_{\text{SFT}}$ is supervised fine-tuning cross-entropy on the gold radiology-report tokens, and $\mathcal{L}_{\text{faith}}$ is the causal-faithfulness loss (§3.4). The faithfulness weight λ_{faith} is a fixed hyperparameter set to 0.3 throughout, chosen by a small smoke-search over $\{0.1, 0.3, 1.0\}$ on a 6-step pre-flight.

3.3. Dual adversarial filter

The dual filter combines two independently-trained shortcut detectors: a text-only branch that scores how predictable the gold label is from the claim and evidence text alone, and an image-only branch that scores how predictable the gold label is from the image alone. Each branch produces a per-row weight in $\{w_{\min}, 1.0\}$ (discrete schedule) or $[w_{\min}, 1.0]$ (continuous SFC ablation). The dual weight is $w(x) = \min(w_{\text{text}}(x), w_{\text{image}}(x))$: a row is downweighted aggressively if it is unimodally solvable along either axis.

The text-only branch reuses the implementation from our prior verifier work [citation withheld for double-

blind review]: a RoBERTa-large cross-encoder trained for one epoch on (claim, evidence) $\rightarrow$ gold-label pairs from the training split. Rows where the text-only model’s predicted probability of the gold class exceeds 0.7 are downweighted to $w_{\min} = 0.2$; the remaining rows are kept at 1.0.

The image-only branch is the genuine novelty axis. We extract image features via the BiomedCLIP image encoder (Zhang et al., 2024) and train a 2-layer MLP head on image $\rightarrow$ gold-label for one epoch. The same threshold rule is applied. On the 20,560-row training pool, the text-only branch downweights 85.3% of rows; the image-only branch 81.2%; the union after min aggregation 91.7%. Crucially, **6.5%** of rows are caught by the image-only branch but missed by the text-only branch (Table ??, App. ??) — exactly the rows whose downweighting cannot be reproduced by a text-only filter alone, and the genuine novelty axis relative to Li et al. (2025a). In an ablation we additionally compare the discrete $\{w_{\min}, 1\}$ rule against the continuous SFC formulation $\text{SFC}(x) = 1 - p_{\text{text}}^{\text{correct}}(x)$ clipped to $[w_{\min}, 1]$.

3.4. Causal-faithfulness loss with token-wise gain margin

The naive consistency loss $\text{ReLU}(\text{margin} - \text{KL}(p_{\text{full}} \| p_{\text{alt}}))$ admits a degenerate optimum: the model can drive p_{alt} to an arbitrary distribution while p_{full} is shaped by the SFT loss alone, and the divergence margin can be satisfied without the model ever using the image. This is the induced inversion failure mode that `IMG_correct` directly measures (§3.5). To rule it out at training time, we replace the symmetric KL with a correctness-gain margin on gold tokens:

$$\mathcal{L}_{\text{faith}}(x) = \frac{1}{|T|} \sum_{t \in T} \text{ReLU}(\text{margin} - (\log p_{\text{full}}(y_t) - \log p_{\text{alt}}(y_t))) \quad (2)$$

where y_t is the gold token at position t , p_{full} and p_{alt} are the per-position distributions under the full and image-zeroed inputs, T is the set of assistant-turn token positions, and $\text{margin} = 0.3 \text{ nats}$ (a deliberately small value; faithfulness-via-reduction-in-gold-token-probability is what we want, not faithfulness-via-arbitrary-distributional-divergence). The token-wise margin couples the divergence requirement to gold-token correctness on the full input, so the model cannot satisfy the margin by being arbitrarily wrong on the image-zeroed input. The same margin acts symmetrically across both image-zeroed and image-flipped counterfactuals; on rows that are not laterality-sensitive, the image-flipped counterfactual is replaced with

the image- zeroed counterfactual as a no-op fallback (masked out via flipped_row_mask).

Image-zeroing is implemented as the per-channel mean of the input batch’s pixel values, not a literal zero tensor: a literal zero is, after the processor’s normalisation, a specific out-of-distribution input that the model has never seen at training time. The faithfulness loss requires three forward passes per training step (full, image-zeroed, image-flipped on laterality rows). To stay memory-bounded on a single H100 we use batch size 2 with gradient-accumulation factor 8, yielding effective batch size 16.

3.5. IMG_correct: methodological refinement of the image-masking gap

The standard image-masking gap is $\text{IMG}(f, \mathcal{D}) = \text{acc}(f, \mathcal{D}) - \text{acc}(f, \mathcal{D}^{\text{img}=0})$. This counts every example on which the model is correct on the full input and incorrect on the image-zeroed input as evidence of image-grounded prediction. But a consistency-loss-trained model can lower its image-zeroed correctness to zero by always flipping its argmax under masking, even if its gold-class probability barely drops. We call the cases where the model flips its argmax under masking but does not drop its gold-class probability by $\geq \tau$ the induced inversion sub-population: observationally identical to faithful grounding under IMG, but mechanistically distinct.

IMG_correct addresses this by counting only those examples where (a) the model is correct on the full input ($\arg \max_{\text{full}} = y$); (b) the model is incorrect on the image-zeroed input ($\arg \max_{\text{masked}} \neq y$); and (c) the gold-class probability under masking is lower than under the full input by at least margin τ : $p_{\text{full}}(y) - p_{\text{masked}}(y) \geq \tau$. We additionally report induced inversion as $(n_{\text{ci,no-margin}} - n_{\text{ci,with-margin}})/n \times 100$, isolating the sub-population of “model flipped under masking but its correct-class probability did not drop by $\geq \tau$.” Note IMG_correct can exceed IMG when many examples transition incorrect-to-correct under masking; this is mathematically why several configurations in our retroactive audit (§5.4) show IMG_correct > IMG. For autoregressive generators we apply the same logic per token on the gold target sequence and average over assistant-turn positions only.

4. Experimental Setup

Data. All training and evaluation use public-data sources. Training distribution: OpenI (Demner-Fushman et al., 2016) — 10,839 (claim, image, gold-report) tuples after our text-target filter. In-

Table 1. Headline IMG_correct on the 1,587-example OpenI test split (token-level, $n = 33,173$). Seed-mean $\pm$ standard deviation over three seeds (42, 1337, 9001) per configuration. Induced inversion is essentially zero across all faith-loss configurations.

Config	IMG_corr (pp)	IMG (pp)	Inv (pp)	Acc (%)
sft_only	2.38 ± 0.26	0.36	0.40	74.31
sft_dual	2.10 ± 0.24	0.89	0.58	75.41
sft_faith	74.11 ± 0.26	74.04	0.01	75.61
sft_full	74.96 ± 0.83	74.96	0.01	75.48
sft_full_sfc	75.42 ± 0.28	75.43	0.01	75.44

distribution evaluation: standard OpenI test split (1,587 examples). Cross-site eval: ChestX-Det10 silver- target subset (1,122 rows) (Liu et al., 2020); PadChest-GR (Castro et al., 2024) is deferred to follow-up because no silver targets are available. We do not use MIMIC-CXR (Johnson et al., 2019) in this study.

Configurations. Five configurations $\times$ three seeds: sft_only (SFT-only baseline); sft_dual (SFT + dual filter, no faith loss); sft_faith (SFT + faith loss, no dual filter); sft_full (full method: SFT + faith + dual filter, the headline configuration); sft_full_sfc (full method with the continuous Li et al. (2025a)-style SFC reweighting). Each runs at seeds {42, 1337, 9001}. Total training cost $\approx$ \$200 on Modal H100 80 GB.

Metrics. Primary: IMG_correct at margin $\tau = 0.1$. Secondary: standard IMG, induced-inversion contribution, accuracy on supervised target tokens. We do not report report-quality metrics (BLEU, ROUGE, BERTScore-style overlap) as primary because the OpenI-only training distribution makes those a single-site comparison rather than a generalisation claim; the central methodological claim is the cross- counterfactual diagnostic.

5. Results

5.1. IMG_correct on the headline five-config ablation

We report IMG_correct, IMG, induced inversion, and accuracy for each of the $5 \times 3 = 15$ Pivot fine-tunes against the SFT-only seed-averaged baseline. The pre-registered prediction is that the full-method configuration achieves IMG_correct strictly greater than SFT-only by at least 5 pp at matched accuracy. All 15 fine-tunes have completed. Numbers in Table 1 are seed-means and standard deviations over the three independent seeds per configuration, aggregated over 33,173 valid assistant-turn token positions across the OpenI test split (≈ 21 tokens/example).

Table 2. Cross-counterfactual eval on sft_full (seed 42 only). Single-seed numbers; the trained-mean cell (74.00 pp) is one realisation of the $\text{IMG_correct} = 74.96 \pm 0.83$ pp distribution from Table 1. Baseline anchor (sft_only seed 42 under crop): $\text{IMG_correct} = 2.80$ pp at 73.82% accuracy.

Counterfactual	IMG (pp)	IMG_corr (pp)	Inv (pp)
mean (TRAINED)	73.97	74.00	0.01
noise	62.10	62.20	0.10
crop	3.64	4.81	0.61
occlude	0.28	1.11	0.48

The pre-registered ≥ 5 pp threshold is exceeded by an order of magnitude on three independent configurations $\times$ three independent seeds each. The full method gives a +72.58 pp improvement at +1.17 pp accuracy over the SFT-only baseline; the continuous-SFC variant gives +73.04 pp at +1.13 pp; the faith-loss-alone configuration gives +71.73 pp at +1.30 pp. Reading down the table, the dual filter alone (sft_dual) is statistically indistinguishable from baseline; the faith loss (sft_faith) is the load-bearing component; adding the dual filter on top of the faith loss contributes a further +0.85 pp seed-mean (within one standard deviation of sft_full); the continuous-SFC variant contributes a further +0.46 pp on top of that. Induced-inversion contributions are essentially zero across all faith-loss configurations (≤ 0.01 pp), meaning the entire IMG gap is faithful grounding under our refined definition.

5.2. Cross-counterfactual eval

The principal limitation of §5.1 is that the faith loss and the IMG_correct metric measure overlapping per-token quantities: a model that minimises the faith loss is mechanically pushed toward higher IMG_correct on the exact counterfactual distribution the loss trained on (per-channel mean image-zeroing). To disambiguate “the model learned its training objective” from “the model genuinely uses the image,” we evaluate the headline sft_full configuration under three additional counterfactuals the faith loss did not see during training: (i) noise replaces pixels with Gaussian noise at the per-channel mean $\pm$ std; (ii) crop keeps only a $30\% \times 30\%$ upper-left patch of the image ($\approx 9\%$ of image area) and replaces the rest with the per-channel mean; (iii) occlude zeroes a centred $50\% \times 50\%$ patch ($\approx 25\%$ of image area) and preserves the periphery.

The result is a clean partial-circularity confirmation. Under the trained counterfactual (mean), IMG_correct is 74.00 pp on this seed (consistent with the 3-seed reading). Under a similar untrained coun-

Table 3. Cross-site eval on ChestX-Det10 silver targets. The sft_full cross-site run was capped at 500 rows; the baseline used the full 1,122-row silver-target subset. IMG_correct is a per-token rate and is not biased by the absolute token count.

Config / Site	n_tok	IMG_corr	Inv	Acc
sft_full / OpenI (3-seed)	99,519	74.96	0.01	75.48
sft_full / chestx (silver, $s=42$)	15,955	54.62	0.03	54.65
sft_only / chestx (silver, $s=42$)	35,944	5.12	0.58	50.33

terfactual (noise: also a globally-uniform-OOO pattern), IMG_correct stays high at 62.20 pp. Under dissimilar untrained counterfactuals (crop, which preserves 9% of true image; occlude, which zeroes the centre), IMG_correct collapses to 4.81 pp and 1.11 pp respectively. The matched-counterfactual sft_only baseline yields $\text{IMG_correct} = 2.80$ pp on crop. The cross-counterfactual $\text{sft_full} - \text{sft_only IMG_correct}$ gap on crop is therefore $4.81 - 2.80 = 2.01$ pp, an order of magnitude smaller than the 72.58 pp gap on the trained-mean counterfactual.

The honest reading: the +72.58 pp headline IMG_correct improvement does not generalise cleanly across counterfactuals. The model has demonstrably learned to drive the gold-token probability to ≈ 0 under the per-channel-mean and noise counterfactuals (which are global, uniform-OOO distributions), but it has not learned a more general “use the image” behaviour that transfers to localised counterfactuals.

5.3. Cross-site transfer (ChestX-Det10 silver)

The OpenI-only training distribution means the headline numbers are in-distribution. To test cross-site transfer we evaluate sft_full and sft_only on a ChestX-Det10 (Liu et al., 2020) silver-target subset generated under our prior radiology-report-generation infrastructure (the silver labels are CheXagent-derived, not radiologist gold; the eval is a transfer test, not a faithfulness-against-radiologist test).

The cross-site sft_full shows $\text{IMG_correct} = 54.62$ pp at 54.65% accuracy (≈ 21 pp drop in both metrics relative to in-distribution). The drop is roughly proportional. The cross-site sft_only baseline yields $\text{IMG_correct} = 5.12$ pp at 50.33% accuracy: it preserves accuracy within 4 pp of in-distribution but barely uses the image. The cross-site $\text{sft_full} - \text{sft_only IMG_correct}$ gap is 49.5 pp, about 68% of the in-distribution gap of 72.58 pp, consistent with partial transfer rather than collapse.

Table 4. IMG_correct retroactive audit (selected cells). Full table at Appendix ???. The pre-registered concern was that $\geq 30\%$ of v5_3_contrast’s IMG would resolve to induced inversion; the empirical induced inversion is 1.1% of IMG.

Config	IMG_corr	IMG	Inv	Inv/IMG
v5_0_base	3.83	1.28	0.17	13.2%
v5_3_contrast	71.05	68.12	0.74	1.1%
v5_4_final	68.43	63.55	0.40	0.6%
v6_0_3site	74.51	70.21	0.07	0.1%
abl_no_consist	3.60	2.79	0.30	10.8%
abl_no_contrast	63.28	60.02	0.87	1.4%
abl_hothresh_80	77.98	74.78	0.74	1.0%

5.4. IMG_correct retroactive audit on prior systems

We apply IMG_correct to the v5 / v6 verifier checkpoints of our prior work [citation withheld for double-blind review] (nine verifier configurations and nine v5 ablations, each on $n = 2,974$ verifier rows). Table 4 summarises the headline cells; the full audit set (18 cells) appears in Appendix ??.

Headline finding (negative result; reported transparently). Across the nine headline configurations and nine v5 ablations, induced-inversion contributions are uniformly small: between 0.00 pp (v6_0_loo_no_openi) and 1.31 pp (v6_0_loo_no_padchest); in relative terms, between 0.0% and 13.2% of the corresponding IMG. The decisive cell — v5_3_contrast, the configuration corresponding to the previously-reported headline IMG ≈ 69 pp claim — yields IMG = 68.12 pp, IMG_correct = 71.05 pp, induced inversion = 0.74 pp (only 1.1% of IMG). The newly-completed v6_0_3site cell — the union-of-three-sites configuration — yields the cleanest reading at full-data scale: induced inversion = 0.07 pp = 0.1% of IMG. The abl_no_consist ablation (consistency loss ablated entirely) is the second-most decisive cell: removing the consistency loss collapses IMG_correct to $\approx$ baseline (3.60 pp) while simultaneously reducing induced inversion (0.30 pp), the opposite of what the pre-registered concern (consistency loss causes induced inversion) would predict. The pre-registered $\geq 30\%$ hypothesis is decisively refuted across the entire audited set.

6. Discussion

What IMG_correct reveals about prior systems. The honest reading of §5.4 is that the IMG metric, as previously reported on the v5/v6 verifier family, was already largely measuring real image-grounded behaviour. The methodological concern that motivated

IMG_correct — that consistency-loss training could in principle reward the model for becoming wrong-under-masking without using the image — remains a real concern theoretically, but the empirical evidence does not support the claim that consistency-loss training produced substantial induced inversion at the calibration tested. The strongest version of the IMG_correct contribution, given this result, is not “audit reveals overstated grounding claims”; it is (a) a verification that prior IMG numbers held up under a strict per-example correctness criterion, increasing confidence in prior claims; and (b) a default-to-report-alongside-IMG diagnostic that gives future workers a way to detect the failure mode if a less calibrated training recipe were to introduce it. We avoid the temptation to relax the pre-registered hypothesis post-hoc.

Cross-counterfactual partial generalisation. §5.2 gives the operational meaning of the circularity caveat: the headline +72.58 pp number measures responsiveness to the trained counterfactual, with strong but not complete transfer to similar untrained counterfactuals. This paper demonstrates that training-time counterfactual interventions can be tightly-fit on the in-distribution counterfactual, not that the resulting model is image-grounded in a deeper sense. A more diverse training-time counterfactual mix (random crop / occlude / noise / mean per training step) is the natural follow-up; we expect it would close the gap.

Cross-site partial transfer. §5.3 shows sft_full’s image-grounded behaviour transfers cross-site (IMG_correct = 54.62 pp on ChestX-Det10 silver, $\approx 68\%$ of in-distribution gap retained) when the counterfactual is held constant (mean-masking) but the test distribution changes. Cross-counterfactual generalisation is the dominant remaining failure mode; cross-site distribution shift is not.

Filter calibration is task-mismatched. The text-only and image-only adversarial filters were trained for the verifier task in our prior work [citation withheld for double-blind review] and reused as shortcut detectors for the present generator task. The conceptually-aligned filter for the new task would ask whether a text-only or image-only generator could reproduce the gold report; our existing filter is a directional proxy. The dual-filter ablation in Table 1 still demonstrates the mechanism, but the calibration is approximate.

Public-data-only training is a feature, not a bug. The reproducibility literature in medical imaging consistently flags access barriers as a constraint on

independent replication of MIMIC-CXR-trained results. Our public-data-only commitment means any researcher with a Hugging Face account and a Modal-or-equivalent compute provider can re-run the entire pipeline. This restriction costs us scale (MIMIC-CXR has $\approx 50\times$ more rows than OpenI) but the cost is recoverable: the methodology generalises to MIMIC-CXR for any researcher with the relevant access, and the results we report should be read as a lower bound on what is achievable with comparable methods on larger training distributions.

7. Limitations

(L1) The faith loss (§3.4) and the IMG_correct metric (§3.5) measure overlapping per-token quantities; the headline +72.58pp number is best read as evidence that the model has fit the training objective on held-out data, not as independent evidence of image-grounded prediction. §5.2 quantifies this caveat empirically. (L2) OpenI-only training; cross-site transfer is measured on ChestX-Det10 silver targets only. PadChest-GR cross-site is deferred. (L3) Filter calibration is task-mismatched (§6). (L4) LoRA rank 16 (11.9M trainable parameters of 4.31B); we have not run a rank ablation. (L5) Single backbone (MedGemma-4B-IT). (L6) Image-masking is one of several plausible counterfactuals; image-cropping or occlusion test different aspects of image-grounded prediction and yield different IMG_correct numbers. We do not claim image-zeroing is the canonical counterfactual; we claim that any counterfactual benefits from the IMG_correct refinement over the IMG metric. (L7) Cross-system retroactive audit is on our own prior verifier checkpoints only; closed-weight commercial vision-language systems and tokenizer-embedded systems (MAIRA-2, CheXagent-2-3B) are deferred because the per-token softmax extraction required for IMG_correct is non-trivial or unavailable under those paradigms. (L8) Single radiology modality (chest radiographs); the methodological contribution generalises to any image-conditioned generation task where text-only and image-only shortcuts are plausible, but we have not tested this.

8. Conclusion

We presented dual-adversarial causal-faithfulness training, a LoRA-based fine-tuning recipe for image-conditioned medical generators that addresses evidence-blindness symmetrically along both modalities and rules out the induced-inversion failure mode at training time via a token-wise correctness-gain margin loss. We additionally introduced IMG_correct,

a methodological refinement of the standard image-masking gap. The pre-registered empirical claims, tested against actual results: (i) Full method reduces IMG_correct on OpenI-test by ≥ 5 pp at matched accuracy: HIT BY AN ORDER OF MAGNITUDE on three seeds $\times$ three faith-loss configs (+72.58pp); (ii) IMG_correct refinement reveals $\geq 30\%$ of v5_3 contrast’s IMG as induced inversion: REFUTED (1.1%); (iii) dual filter dominates the faith loss in the ablation: INVERTED (the faith loss is the load-bearing component). One pre-registration HIT, one REFUTED, one INVERTED, all reported transparently. The methodological refinement (IMG_correct) lands as a contribution independently of the empirical results, since it is a reusable audit-tool that improves the interpretation of any future or prior IMG-style metric. We advocate that counterfactual evidence-sensitivity testing precede aggregate-accuracy reporting for any multimodal medical generator, and that the IMG_correct margin become a default reporting requirement.

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